Columbia Business School

B9339 Systematic Investment Strategies
Professor: Achilles Venetoulias
Fall 2026

Homework Assignment #3

Due: Thursday, October 15, 2026, 2:20pm (EST)

- Homework assignments can be done in groups of three or fewer people. Only turn in one write-up per group.

- In your write-up, please:
  - Include (preferably on the first page) a summary of your answers to all the questions.
  - Give your answers to at least five digits (if appropriate) after the decimal place (this makes it easier to spot which answers are correct or where mistakes occur).

- In your solutions to the individual questions, please:
  - Explain the main steps or formulas used to arrive at your solution. Specifically, *show* and *explain* the main formulas used in any spreadsheet or software output you hand in. The formulas in the annotated output can be handwritten (do not have to be typeset).
  - Include row and column headings in spreadsheet printouts (in Excel, choose *File—Print Preview*, then click on *Setup*, then choose the *Sheet* tab and check the "row and column headings" box).

- Please organize and structure software output you hand in so that it is easy for others to understand. Spend extra time, after getting the answers correct, to create clean-looking, understandable submissions.

- It is not sufficient to turn in a spreadsheet or some other software output (in printout or electronic form) with only numbers showing. Points will be deducted for write-ups turned in without the main formulas shown and explained, even if the numerical solutions are correct. How much detail is needed in the explanations? Enough detail so that anyone else in the course (another student, the TA, the instructor) could follow (and, relatively easily, reproduce) your work and final solutions using only your write-up.

- Late homework submissions will not be accepted.

1) (*FX carry*) In this assignment, you are to investigate FX carry strategies. The Excel spreadsheet "hw3.data.FX.dly.xlsx" contains daily data:
   - FX spot rates for the G-10 currencies: EUR, JPY, GBP, CAD, AUD, NZD, CHF, NOK, SEK (columns B-J). All exchange spot rates are against the US dollar (USD), and they are quoted in their standard market convention: the "EUR" quote is USD/EUR but the "JPY" quote is JPY/USD
   - 1-day returns for each spot rate (columns L-T)
   - Interest rates for the G-10 countries (columns W-AP). The interest rates correspond to 1-month rates (labeled as "… 1M") and 3-month rates (labeled as "… 3M")
   - Implied volatilities (from the currency options market) for:
     - the AUD/JPY spot rate at tenors of 1 week (column AR: "AY1wk"), 3 months (column AS: "AY3M"), and 6 months (column AT: "AY6M");
     - for JPY/USD at tenor of 3 months (column AU: "JPY3M");
     - for AUD/USD at tenor of 3 months (column AV: "AUD3M")
   - The VIX index (aka the "fear index", constantly mentioned on CNBC), which is the Chicago Board Options Exchange's volatility index and reflects the market's *forward* expectation of 30-day equity volatility (hence the moniker "fear index").

   Use the FX spot and interest rate data to construct carry portfolios that:
   i. Are based on the USD-based basket of currencies;
   ii. Are based on the 1-month and the 3-month rates;
   iii. Contain 2 and 3 currencies each on both the long and the short sides (for a total of 4 or 6 positions respectively);
   iv. Use an equal, a rank-based, and a proportional-to-carry allocation for each position in the portfolio.

   (a) For each of the 12 portfolios you have constructed (2 rates X 2 numbers of currencies X 3 allocation methods), clearly state the currencies you are using and the weights of each position in the portfolio; also, explain how you have arrived at these choices.

   (b) Compare the distributions of the return streams of your carry portfolios. Does one construction jump out as "better" than the others? Is there a single factor in your construction (i.e., rate tenor, number of currencies included, allocation method) that appears to make a bigger difference than the others? Briefly explain if you think your findings are intuitive or not.
   [Note: (i) To "compare" the distributions, it is not enough to say one portfolio produces more return than another portfolio. You need to compare arithmetic and geometric returns, other moments of the distributions, drawdowns, and any other properties of the distributions you think are relevant. (ii) The explanation on the intuition is an open-ended question: there is no right or wrong answer, as long as your explanation makes sense.]

   (c) Analyze the relationship of the "best" carry portfolio you have identified in (b) to the volatility data in your dataset. Which volatility time-series seems the be the most useful candidate for a conditioning variable that may help you improve your carry portfolio, and why? Does your finding make intuitive sense (in terms of underlying assets, volatility tenor, etc.)? In particular, do the data confirm (for your carry portfolio) the relationship between FX carry, FX volatility, and equity volatility that Jurek describes?

(d) Apply a conditioning scheme of your choice to each of the volatility data variables to improve upon the "best" FX carry portfolio you have created in (c). Which of the resulting 6 carry portfolios emerges as the best one? Is this answer in agreement with (i) what you would have expected on the basis of your analysis in (c) and (ii) your intuition?
[Note: (i) You may choose the same or a different conditioning strategy for each volatility variable, and you do not need (in fact, you should not attempt) to 'optimize' your conditioning strategies. (ii) In your answer viz-a-viz (c) you should explain why this question asks you to look at all the volatility variables and not just at the candidate (volatility variable) you have identified in (c).]

(e) Choose one (1) measure of historical volatility of your choice (e.g. historical volatility of a particular spot rate, historical volatility of a basket of spot rates, historical volatility of one of the implied volatilities, historical volatility of the equity market, etc.) and repeat step (d). Compare the return distribution of this carry portfolio (conditional on the measure of historical volatility you chose) to the return distribution of the "best" carry portfolio (conditional on implied volatility) in step (d). Explain if/why your findings agree with your intuition.
[Note: (i) For this step, you need to explain the motivation behind the measure of historical volatility you are choosing; again, there is no right or wrong answer, as long as your explanation is clear and reasonable. (ii) If you choose to use historical volatility of the equity market(s), either in the US or worldwide, you will need to download the appropriate/necessary data; Yahoo Finance and/or Bloomberg are convenient sources that provide such data with relatively little effort.]

For the purposes of this assignment:

- All "returns" are to be computed as arithmetic returns:

$$\frac{P_{t+1}}{P_t} - 1$$

- Transaction and other costs are to be ignored
- Rebalancing of the strategies is to be done on a daily basis (same time scale as the data provided)